

Reflective Report

This project gave me hands-on experience across the full data pipeline. I was involved from early on, contributing to data cleaning and merging, which turned out to be more demanding than expected. The data was messier than we anticipated, and the cleaning and merging stages had to be revisited more than once to get reliable outputs. This taught me that early-stage data work is not something to rush, because problems there carry through into everything that follows.

Beyond data preparation, I worked across the EDA and modelling stages. The EDA was particularly valuable because it revealed how complex the relationships between process variables and capsule quality actually were. This shaped the direction the team took with modelling and helped narrow down what mattered most.

I also took on the role of communicating our progress and findings to Albotherm on behalf of the team. This pushed me to translate technical work into clear, accessible language, which is a skill I developed noticeably over the course of the project.

If I were to change one thing, it would be investing more time early on in understanding the domain and chemistry behind the process. Having that grounding earlier would have shaped my interpretation of the data faster and more confidently.

Overall, this project strengthened my ability to work with real, imperfect data, contribute meaningfully across different stages of analysis, and communicate findings clearly. These are skills I expect to carry into future work.